IGCSE 0607 Practice Questions — E2.4

Indices II

Total Questions: 33 — **Total Marks:** 186 Marks

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  → **Paper 2 (Non-Calculator)** *(25 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

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1 Simplify fully.

$4x^3y^{-2} \times 3x^5y^6$

..... **[2]**

2 **(a)** Simplify.

$$4x^2 \times 3x^{-5}$$

..... **[1]**

(b) Simplify.

$$\left(\frac{8x^9}{27}\right)^{\frac{1}{3}}$$

..... **[2]**

3 **(a)** Simplify.

$4x^3 \times 5x^{-2}$

..... **[1]**

(b) Simplify.

$$\left(\frac{2x^4}{y^3}\right)^2$$

..... **[2]**

(c) Solve the equation.

$$3^p = \frac{1}{27}$$

$p =$ ..... **[1]**

4 **(a)** Simplify.

$$4x^3y^{-2} \times 3x^5y^6$$

..... **[2]**

(b) Find the exact value of

$$16^{-\frac{1}{4}}$$

..... **[2]**

5 **(a)** Simplify.

$5x^4 \times 3x^{-2}$

..... **[2]**

(b) Find the value of

$16^{\frac{3}{4}}$

..... **[2]**

6 **(a)** Simplify.

$y^5 \times y^{-2}$

..... **[1]**

(b) Work out the value of

5^{-2}

..... **[1]**

(c) Simplify.

$(2x^4)^3$

..... **[2]**

7 **(a)** Simplify

$$4x^{-3} \times \frac{3}{2}x^{\frac{1}{2}}$$

..... **[2]**

(b) Simplify

$$\left(\frac{3y^4}{2}\right)^3$$

..... **[2]**

8 **(a)** Find the value of x when

$$2^x = \frac{1}{8}$$

$x =$ ..... **[1]**

(b) Simplify

$$(4y^6)^{\frac{1}{2}}$$

..... **[2]**

(c) Work out the exact value of

$$16^{-\frac{3}{4}}$$

..... **[1]**

9 **(a)** Simplify.

$w^4 \times w^7$

..... **[1]**

(b) Simplify.

$(3x^3)^2$

..... **[2]**

(c) Simplify.

$$\frac{15y^8}{5y^{-2}}$$

..... **[2]**

10 **(a)** Simplify.

$p^4 \times p^7$

..... **[1]**

(b) Simplify.

$(3q^3)^2$

..... **[2]**

(c) Work out the value of $27^{\frac{2}{3}}$.

..... **[2]**

11 **(a)** Simplify fully

$$\frac{12x^5y^3}{3x^2y^{-1}}$$

..... **[2]**

(b) Solve the equation

$$27^{x-1} = 9^{2x+1}$$

$x =$ ..... **[3]**

12 **(a)** Work out the exact value of

$$32^{-\frac{2}{5}}$$

..... **[2]**

(b) Solve the equation.

$$5^{x+2} = 125^{x}$$

$x =$ ..... **[3]**

13 **(a)** Work out the exact value of

$$125^{\frac{2}{3}}$$

..... **[2]**

(b) Simplify fully.

$$\frac{15x^4y^{-3}}{5x^{-2}y^5}$$

..... **[3]**

14 **(a)** Solve the equation.

$16^x = 2$

$x =$ ..... **[1]**

(b) Work out the exact value of

$125^{\frac{2}{3}} \times 5^{-2}$

..... **[2]**

(c) Solve the equation.

$5^{2x-1} = 125^x$

$x =$ ..... **[3]**

15 **(a)** Simplify fully.

$$4x^{-3} \times \frac{3}{2}x^{\frac{1}{2}}$$

..... **[2]**

(b) Solve the equation.

$$3^{2x+1} = 27$$

$x =$ ..... **[2]**

(c) Simplify fully.

$$\left(\frac{8a^6}{b^3}\right)^{\frac{1}{3}}$$

..... **[2]**

16 **(a)** Simplify fully.

$$\left(\frac{2x^5}{3y^2}\right)^3$$

..... **[3]**

(b) Solve the equation.

$$9^x = 27^{x+1}$$

$x =$ ..... **[3]**

17 **(a)** Work out the value of

$$\left(\frac{27}{64}\right)^{-\frac{2}{3}}$$

..... **[2]**

(b) Solve the equation

$$8^{2x-1} = 32^{x+2}$$

$x =$ ..... **[4]**

18 **(a)** Simplify fully.

$$\left(\frac{27x^9}{64y^{12}}\right)^{-\frac{1}{3}}$$

..... **[3]**

(b) Solve the equation.

$$2^{3x-1} = 32^x$$

$x =$ ..... **[3]**

19 **(a)** Simplify fully.

$$\left(\frac{16x^{12}}{y^8}\right)^{\frac{3}{4}}$$

..... **[3]**

(b) Solve the equation.

$$2^{x+3} = 8^x$$

$x =$ ..... **[3]**

20 **(a)** Solve the equation.

$$9^{2x} = 27^{x+1}$$

$x =$ ..... **[3]**

(b) Simplify fully.

$$\left(\frac{16a^8}{81b^{12}}\right)^{-\frac{3}{4}}$$

..... **[3]**

21 **(a)** Solve the equation.

$$16^{2x-1} = 8^{x+2}$$

$x =$ ..... **[3]**

(b) Simplify fully.

$$\left(\frac{81x^{12}}{16y^8}\right)^{-\frac{3}{4}}$$

..... **[3]**

22 **(a)** Simplify fully.

$$\frac{16x^{\frac{5}{2}}y^{-2}}{2x^{\frac{1}{2}}y^{3}} \div \frac{4x^{3}}{y^{4}}$$

..... **[3]**

(b) Solve the equation.

$$5^{2x-1} \times 25^{x+2} = 125^{x}$$

$x =$ ..... **[4]**

23 **(a)** Simplify fully.

$$\frac{4x^{-\frac{1}{2}} \times 3x^2}{6x^{\frac{1}{3}}}$$

..... **[3]**

(b) Solve the equation.

$$9^{2x-1} \times 3^{x+1} = 27^x$$

$x =$ ..... **[4]**

24 **(a)** Simplify fully

$$\frac{4x^{\frac{3}{2}} \times (9x^{-4})^{\frac{1}{2}}}{6x^{-\frac{3}{2}}}$$

..... **[4]**

(b) Given that $\dfrac{3^{2k-1}}{27^k} = \sqrt{3}$, find the value of k.

$k =$ ..... **[3]**

25 **(a)** Simplify fully.

$$\frac{15x^{\frac{2}{3}} \times 2x^{-\frac{1}{6}}}{5x^{\frac{3}{2}}}$$

..... **[3]**

(b) Solve the equation.

$$x^{-\frac{2}{3}} = \frac{4}{9}$$

$x =$ ..... **[3]**

(c) Solve the equation.

$$9^x \times 3^{2x-1} = 27^{x+1}$$

$x =$ ..... **[4]**

26 **(a)** Simplify.

$5x^2y^3 \times 2x^4y^{-1}$

..... **[2]**

(b) Simplify.

$\left(\frac{27a^6}{b^9}\right)^{\frac{1}{3}}$

..... **[2]**

(c) Evaluate.

$16^{-\frac{1}{2}}$

..... **[1]**

27 **(a)** Simplify fully.

$$\left(\frac{16x^{12}}{y^8}\right)^{\frac{3}{4}}$$

..... **[3]**

(b) Solve the equation.

$$4^{2x-1} = 8^{x+2}$$

$x =$ ..... **[3]**

28 **(a)** Solve the equation.

$$4^{2x-1} = 32^x$$

$x =$ ..... **[3]**

(b) Simplify.

$$\frac{12x^5y^{-3}}{3x^2y}$$

..... **[2]**

(c) Solve the equation.

$$x^{\frac{3}{2}} = 64$$

$x =$ ..... **[2]**

29 **(a)** Simplify fully.

$$\frac{24x^{\frac{3}{2}}y^{-2}}{6x^{-\frac{1}{2}}y}$$

..... **[2]**

(b) Solve the equation.

$$9^x \times 27^{x-1} = 3\sqrt{3}$$

$x =$ ..... **[4]**

30 **(a)** Solve the equation.

$$4^x \times 8^{2x-1} = 16$$

$x =$ ..... **[3]**

(b) Simplify fully.

$$\frac{12x^{\frac{3}{2}} \times 2x^{-\frac{1}{2}}}{8x^{-3}}$$

..... **[3]**

31 **(a)** Simplify

$$\left(\frac{8x^9}{27y^6}\right)^{-\frac{2}{3}}$$

..... **[3]**

(b) Find the values of a and b given that

$$\frac{5^{a-2} \times 25^b}{125^{a+1}} = 5^4$$

$a =$ ..... , $b =$ ..... **[4]**

32 **(a)** Simplify fully.

$$\frac{12x^{\frac{3}{2}}y^{-2}}{4x^{-\frac{1}{2}}y^{4}}$$

..... **[3]**

(b) Solve the equation.

$$4^{2x-1} \times 8^{x+1} = 32^{x}$$

$x =$ ..... **[4]**

33 **(a)** Simplify fully.

$$\frac{18a^3b^{-2}}{45a^{-1}b^4} \times \left(\frac{a^2b}{2}\right)^{-2}$$

..... **[4]**

(b) Solve the simultaneous equations.

$$9^x \times 3^y = 2187$$
$$\frac{8^x}{4^y} = 128$$

$x =$ ..... , $y =$ ..... **[5]**